NeoLAB

Company and Technology Introduction

NeoLAB Convergence Inc

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DIGITAL PAPER neo.LAB convergence

About US

NeoLAB Convergence Inc.

C.E.O

Steve Lee

Established Date

Feb 19. 2009

Employees

65

We are headquartered in Seoul, Korea.

A technology-driven service company composed of multidisciplinary talents with experience in AI, hardware, software, and publishing copyrights.

CEO

- Board Of Directors
- Auditor

Sales & Marketing

- Domestic Business Team
- International Business Team
- Marketing
- Service Planning Team

R&D Team

- Firmware Development Team
- Hardware Development Team
- Software Development Team

Production/CS/AS

- Production & Procurement Team
- Quality Control Team
- Development Quality Team
- Customer Support Team

Business Support Center

- Finance & Accounting
- Human Resources

Area of Business

Technology based on
Ncode and Opto electronics

Smartpen

Enabling paper to capture handwriting

Sound pen

Enabling paper to be interactive

Partners and clients

Technology

Microsoft
USI Universal Stylus Initiative
wacom
KENT DISPLAYS
NTT DATA Global IT Innovator

Consumer & Stationery

MOLESKINE®
LAMY
YANGJISA
monami®
ZEQUENZ

Education | Educational Institute

HYBE EDU
SCHOLASTIC
구몬학습
DAEKYO 대교
SMARTSTUDY
woongjin 웅진씽크빅
megastudy

Enterprise Document Processing

LG화학
HYUNDAI
현대해상

Government / Public Sector

U.S. Department of Veterans Affairs
ひと、くらし、みらいのために 厚生労働省 Ministry of Health, Labour and Welfare
충청남도교육청 CHUNGCHEONGNAMDO OFFICE OF EDUCATION
Department for Education

Ncode - Digital enabler

Digital readiness

Ncode the digitizer

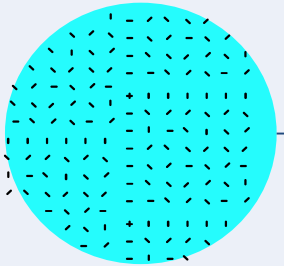
NCODE patterns on each pages will make paper notebooks digital ready

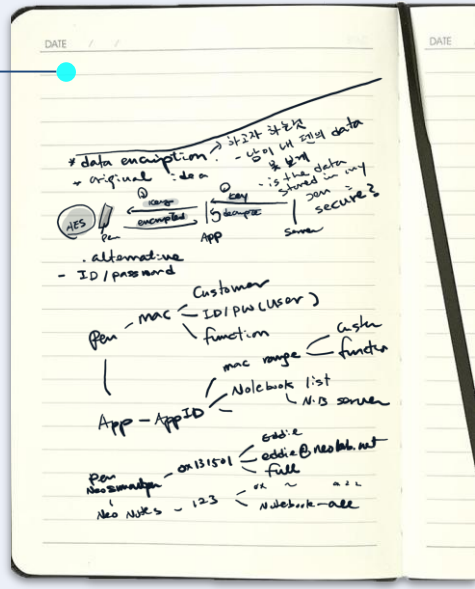
Metadata

It is not only capturing your notes but know what kind of notebook you are writing on and which page you are writing into.

NCODE includes

- Surface Type ID
- Book ID
- Page Number

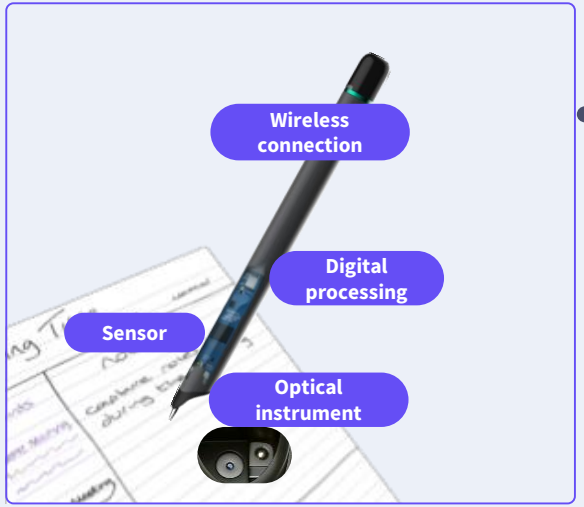




Smartpen solution

Utilizations in diverse industrial fields with a Ncode-reading digital pen

Ncode recognition



Ncode to digital data

An example of the written data

Written/Meta data

Surface data (Publisher, book, page, form ID)

Coordinates (x, y)

Atypical data (Input pressure, pen inclination, rotation)

Visual data (Per millisecond)

Data utilization

Data collection
- Data store
- Data sync

+

Stroke restoration
- Equivalent data restore
- Format switch

+

Data process
- Data study
- Transcription

+

Data utilization
- Customization
- Studied data

Applicable fields

Education	Chemical facilities	Office	Medical
• Correcting guidance • Academic achievement • Orthography • Video/AI ties	• Datafying inspection data • Real-time data investigation • Paper to digital format • Soft copy of inspection report	• Note-taking for a digitized soft-copy • Format per need • Digital whiteboard solution	• Medical history in digital format • Cognitive ability tests • Disgraphia test

Neo smartpen

An optical smartpen that capture your notes, sketches on Ncode paper.

Ink system: Standard D1 Ballpoint, Oil
Pressure level: 1024
Operation time: 6 hours

Ink system: Custom Ballpoint, Oil, Gel.
Pressure level: 4096
Operation time: 11 hours

Moleskine - Smart Pen

A 3rd generation Moleskine Smart pen

LAMY safari all black Ncode

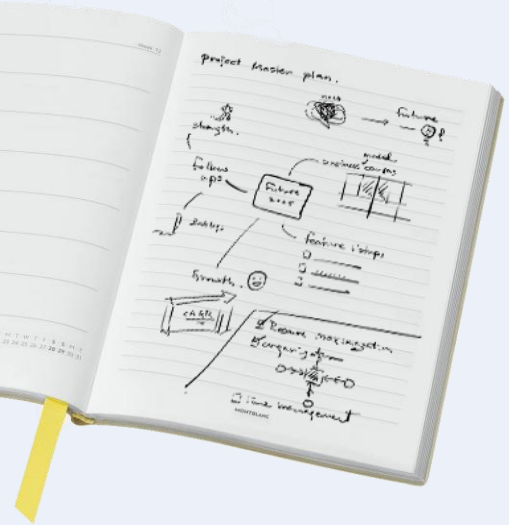
Lamy's legendary safari is transformed to a Smartpen

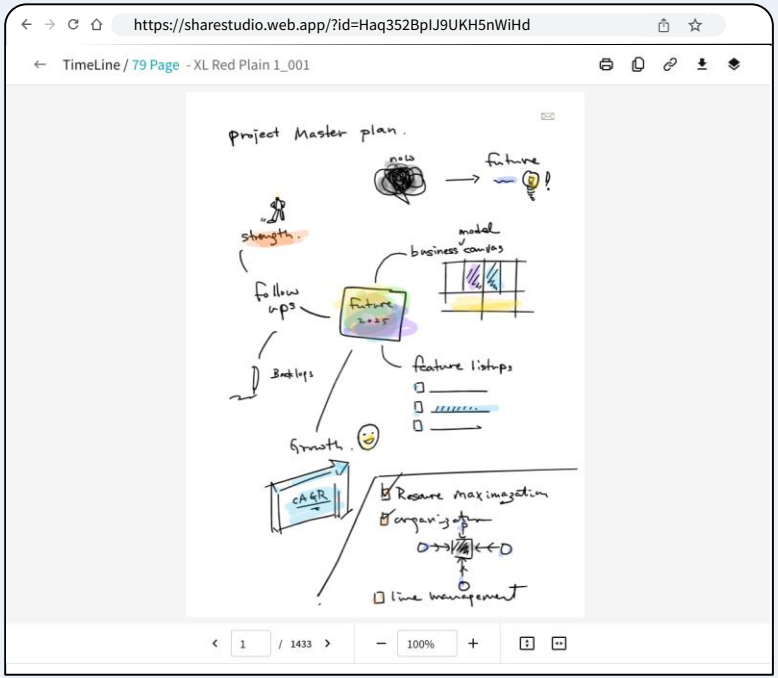
MONAMI 153

A special edition of MONAMI 153

every page has it own internet address

A page can be shared as a URL link. it means you can browser user page from internet whenever you want to refer.

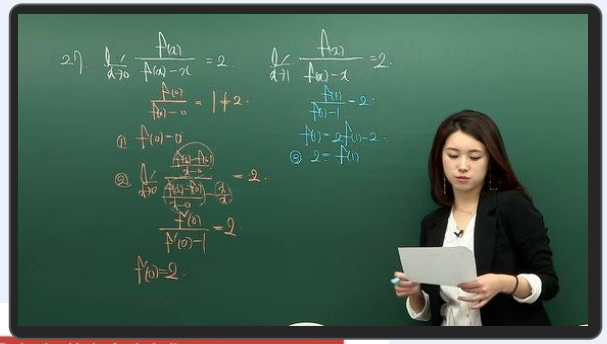


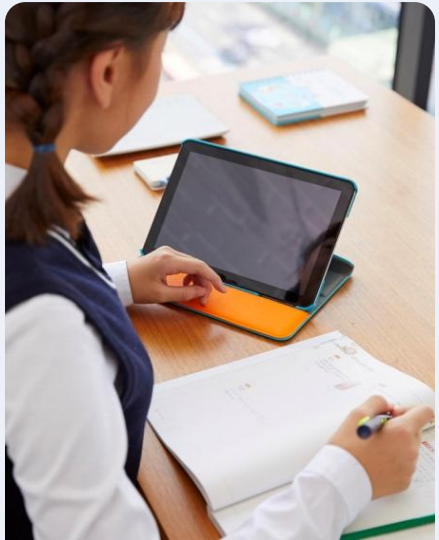


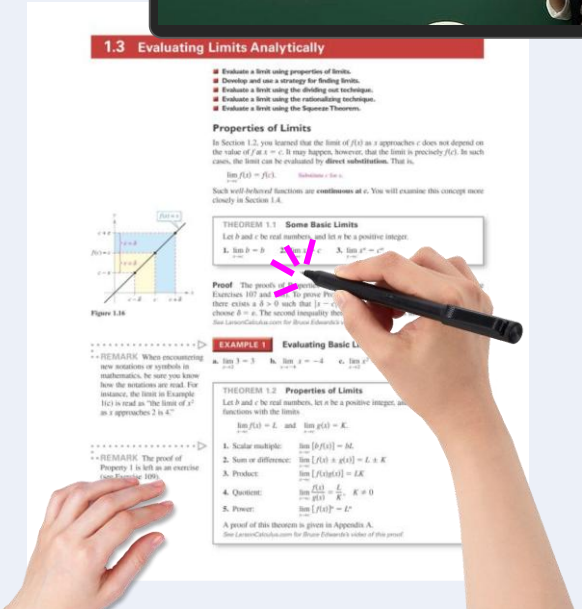
Internet link : https://sharestudio.web.app/?id=Haq352BpIJ9UKH5nWiHd

Trigger lecture video from paper textbook

Paper can have hyperlinks to evoke .







Make a paper planner functionable

Sync to digital calendars

A particular part of page can be defined to sync with digital calendars. Your memos, appointments and events on a paper diary sync to your phone and laptop.

AI(Artificial Intelligence) Assistant

Thanks to the modern AI advancement, the meeting minutes can be summarized and reported via Email

Google
Natural Language

Amazon
Comprehend

Meeting Outcome : 9:30 - 11:00 AM inbox X

Neostudio <neostudio@neolab.net>
to me ▼

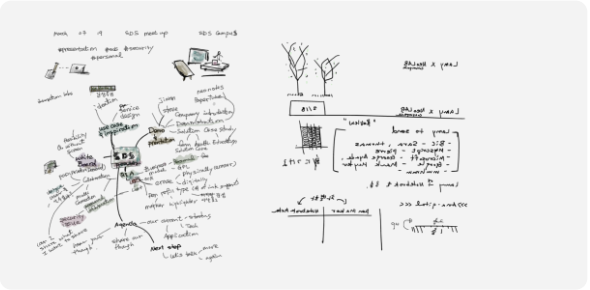
STUDIO

This is a meeting outcome started at 9:30AM

Key takeaways

IDEA DELEGATION TEAMWORK
DENSITY MOTIVE MOCKUP

The actual memo during the meeting



Please check the below link for the detailed content
https://sharestudio.web.app/?id=Haq352BpIJ9UKH5nWiHd

[Meeting outcome email]

Sound Pen

SCHOLASTIC

Scholastic Corporation is an American multinational publishing, education and media company that publishes and distributes comics, books and educational materials for schools, parents and children

Scholastic's reading book adopts NeoLAB's Ncode and those enabled books are interactive with Sound pen from NeoLAB. Teamed up with Scholastic, we are expanding service territories

BTS

BTS is a septet boy band from S. Korea. They are one of the most influential bands of all time and have a fan base of more than 20 million followers.

HYBE EDU x NeoLAB

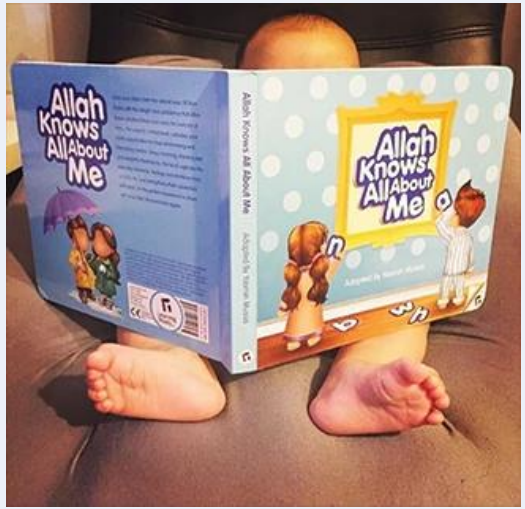
Learn Korean with BTS

In Learning Korean with BTS, the entire book is printed with NeoLAB's proprietary Ncode, and the package comes with a smartpen with a built-in speaker that can interact with a paper book

Learning Roots

Learning Roots have been publishing books for Islamic education since 2004, and selling their products to more than 33 countries worldwide using our sound pen.

Click here to play a video



Recent Developments

POKORO – AI Sound Pen

Target Consumer

- Children aged 5–12 years
- Parents seeking low-screen educational tools
- Schools & libraries promoting interactive reading
- Publishers adopting AI-enhanced book experiences

Key Features & Functions
- *Conversational AI Tutor*
- *Screen-Free Voice Interaction*
- *Personalized Learning Analytics*
- *Multilingual Support: Supports multiple languages for global learning*

Smart Eraser

A digital eraser designed to seamlessly integrate with the Neo Smartpen series. It allows users to **erase both physical and digital ink strokes** simultaneously, enhancing the note-taking experience

Main Features

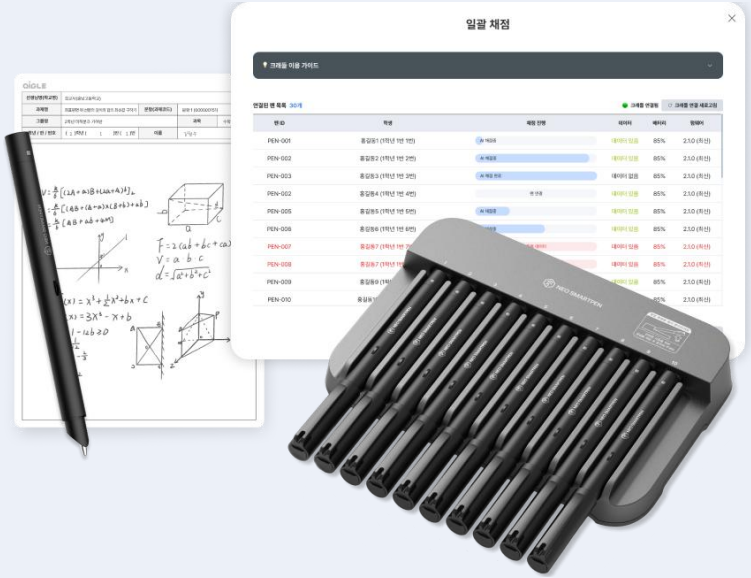
- *Dual Erasing (paper & digital)*
- *D1 Eraser Pen Refills*
- *Type-C Fast Charging*
- *Long Battery Life (up to 40 hours)*
- *Bluetooth 4.2 (BLE) Connectivity*
- *Compact & Lightweight (133mm × 20mm, 35g)*
- *Optimized for Neo Smartpen + Neo Studio 2 app*
- *Real-time synchronized editing*

AI Auto-Grading · Handwriting Recognition · Personalized Feedback · Essay & Short-Answer Evaluation

Aigle is an **AI-powered grading and feedback service** designed for tutors, private academies and school teachers who assess essay and short-answer responses. Students write naturally on paper, and Aigle's AI automatically converts their handwriting into digital text, applies consistent grading criteria, and delivers real-time, individualized feedback — eliminating the subjectivity and time burden of manual grading

Key Benefits:

- *Cuts grading time, freeing teachers for actual instruction*
- *Fair, consistent evaluation with AI-driven criteria*
- *Real-time, personalized feedback per student*
- *Progress tracking by semester and year*
- *Works with handwritten input — no keyboard or tablet needed*



Cradle: Multi-Pen Charging Dock · Classroom Management · Batch Sync

Access service

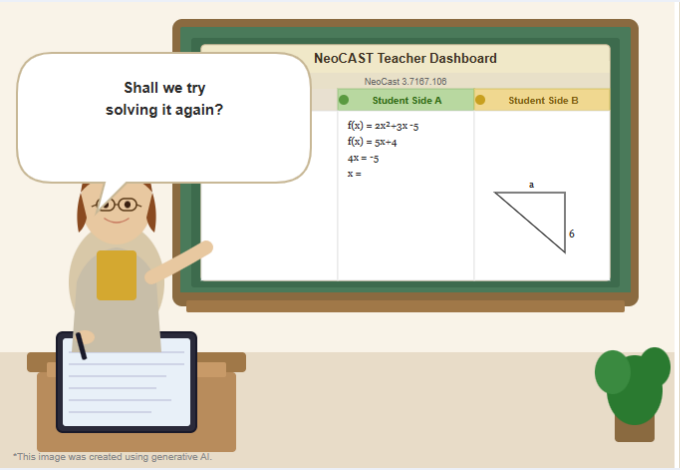
NeoCAST

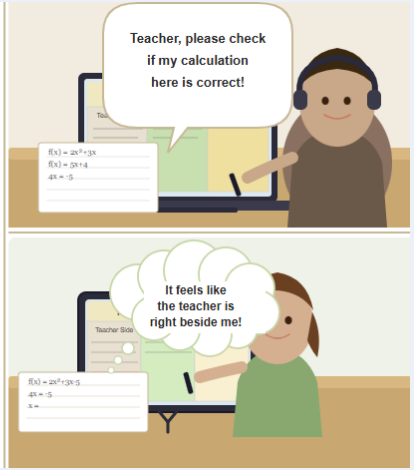
Live Broadcast · Real-Time Handwriting Sharing · Interactive Teaching · No App Required

NeoCAST is a **two-way interactive teaching platform** for tutors and classroom teachers that lets students write naturally on paper while their work is instantly shared on the teacher's screen in real time. Teachers can monitor every student's progress at a glance, identify who needs help, and provide timely, personalized guidance — all without any electronic whiteboard or additional classroom infrastructure.

Key Benefits:

- *See every student's handwriting live via a monitoring dashboard*
- *Deliver 1-on-1 level attention even in group settings*
- *Students join via a simple web link — no app install needed*
- *Voice chat built in, no separate video tool required*
- *All sessions auto-saved for absentees and lesson review*





Access service(trial)

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